Zulip Follow User Feature

Handover Package

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Cornell University

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Repository	https://github.com/Pahina0/zulip
License	Apache 2.0

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1. Project Overview

1.1 Problem Statement

Zulip is an open-source team communication platform used by companies and open-source communities worldwide. While Zulip allows users to follow individual topics, there was no way to follow a specific person so that their messages surfaced more prominently. High-volume workspaces suffer from signal-to-noise problems: important messages from key collaborators are buried in channel feeds.

This project adds a Follow User feature to Zulip, allowing users to designate individuals as followed and receive dedicated push notifications when those users send messages.

1.2 Feature Summary

- Follow / unfollow any user via the right-sidebar buddy list
- Following Feed — a new left-sidebar view showing only messages from followed users
- is:followed-user narrow filter for use in the search bar
- Backend push notification pipeline: followed_user_push_user_ids delivered via the existing Zulip push-notification infrastructure
- Cache-invalidation-aware storage of follow relationships in the FollowedUser model

1.3 Feature Demo

- Demo found at this link: https://www.youtube.com/watch?v=Hr7Z_SgvxBQ

1.4 Scope and Deliverables

Deliverable	Status
FollowedUser database model	Complete — migration applied
Follow/unfollow REST API	Complete — POST/DELETE /api/v1/users/me/followed_users/{user_id}
Follow button (buddy list UI)	Complete — appears on hover, toggles follow state
Following Feed sidebar view	Complete — new left-sidebar entry, is:followed-user narrow
Push notification pipeline	Complete (backend) — enable_followed_user_push_notifications flag; UI toggle pending
Real-time event propagation	Complete — followed_users event type, Tornado/client sync
Unit tests (backend)	In progress — test file drafted, see Section 4
Unit tests (frontend)	In progress — test file drafted, see Section 4
Help-center documentation	Stub created — requires copy-editing before merge

2. User's Manual

2.1 Following and Unfollowing a User

The follow button appears in the right sidebar (buddy list) when you hover over any user's name.

To follow a user:

- Open the right sidebar (click the people icon or press W).
- Hover over the name of the user you want to follow.
- Click the follow button (person-with-plus icon) that appears to the right of their name.
- The button changes to a filled icon confirming you are now following them.

To unfollow a user:

- Hover over the same user in the buddy list.
- Click the unfollow button (person-with-minus icon).

2.2 The Following Feed

The Following Feed is a dedicated view in the left sidebar that displays only messages sent by users you follow. It works like any other Zulip narrow — you can reply, react, and navigate as normal.

To open the Following Feed:

- Click Following feed in the left sidebar under the Views section.

The sidebar entry shows an unread count badge when followed users have sent new messages you haven't read yet.

2.3 Searching for Messages from Followed Users

You can use the search bar to filter messages from followed users in any context:

```
is:followed-user
```

This narrow can be combined with other operators:

```
is:followed-user channel:general
is:followed-user is:unread
```

The search bar will display "messages from users you follow" as the description when this filter is active.

2.4 Privacy Considerations

- Follow relationships are private — followed users are not notified that you follow them.
- The list of users you follow is not visible to other users or administrators through the UI.
- You can follow deactivated users; their past messages will still appear in your Following Feed.
- Bots can be followed the same as regular users.

2.5 Push Notifications from Followed Users

When enabled, you receive a mobile push notification whenever a user you follow sends a message that you can see. This setting is controlled by the enable_followed_user_push_notifications flag on your user profile.

Note: *The UI toggle for this setting has not yet been added to the Notifications settings page. Currently, the flag must be set programmatically. This is the primary remaining work item for a future contributor.*

3. Maintainer's Manual

3.1 Architecture Overview

The feature follows Zulip's standard layered architecture:

Layer	Key Files	Responsibility
Database	zerver/models/users.pyzerver/migrations/	FollowedUser model, enable_followed_user_push_notifications field
Library	zerver/lib/followed_users.py	Query helpers: get_user_follows, get_following_users, get_follow_object
Actions	zerver/actions/followed_users.py	do_follow_user, do_unfollow_user — business logic + event emission
Views (API)	zerver/views/followed_users.py	REST endpoints: follow_user, unfollow_user, get_followed_users
URL routing	zproject/urls.py	Maps /api/v1/users/me/followed_users/... to view functions
Push pipeline	zerver/actions/message_send.py	get_followers_who_can_see_stream_message — computes followed_user_push_user_ids
Caching	zerver/lib/cache.py	get_followed_users_cache_key — cache key for per-user follow sets
Frontend data	web/src/followed_users_data.ts	In-memory Set of followed user IDs; add/remove/set/initialize
Frontend filter	web/src/filter.ts	is:followed-user predicate, metadata (title, icon, description, link)
Frontend nav	web/src/navigation_views.ts	Following Feed sidebar entry, tooltip template reference
Templates	web/templates/tooltip_templates.hbsweb/templates/search_description.hbs	following-feed-tooltip-template, is:followed-user search description

3.2 Database Model

FollowedUser (zerver/models/users.py)

Field	Type	Description
user_profile	FK(UserProfile)	The user who is doing the following
followed_user	FK(UserProfile)	The user being followed
date_followed	DateTimeField	Timestamp when the follow was created (auto_now_add)

enable_followed_user_push_notifications (UserProfile)

A BooleanField (default=False) added to the UserProfile model. When True for a user, they receive push notifications whenever a user they follow sends a message visible to them.

3.3 REST API Endpoints

Method	URL	Auth	Description
GET	/api/v1/users/me/followed_users	Authenticated user	Returns list of followed user IDs with timestamps
POST	/api/v1/users/me/followed_users/{user_id}	Authenticated user	Follow user_id
DELETE	/api/v1/users/me/followed_users/{user_id}	Authenticated user	Unfollow user_id

Error responses:

- 400 You cannot follow yourself.
- 400 User is already followed. (POST to already-followed user)
- 400 User is not followed. (DELETE with no prior follow)

3.4 Actions Layer

do_follow_user(user_profile, target_user)

- Creates a FollowedUser row.
- Invalidates the get_followed_users_cache_key for target_user.
- Emits a followed_users Tornado event so all of the follower's clients update immediately.

do_unfollow_user(user_profile, target_user)

- Idempotent: no-ops (but still emits an event) if the relationship doesn't exist.
- Deletes the FollowedUser row.
- Invalidates cache and emits event as above.

3.5 Caching

get_followed_users_cache_key(user_id) (zerver/lib/cache.py, line ~559) stores the set of user IDs who follow a given user. It is populated lazily by get_following_users(user_id) and invalidated on every follow or unfollow action. Tests must account for this: assert the cache key is None after an action and repopulated after the next call.

3.6 Narrow / Filter Integration

The is:followed-user narrow is handled in web/src/filter.ts:

- Predicate: checks followed_users_data.is_user_followed(message.sender_id).
- Metadata: title = "Following feed", icon = "follow", description = "Messages from users you follow.", link = "/help/follow-a-user".
- can_newly_match_moved_messages: returns true (moved messages from followed users can appear).

The search bar description ("messages from users you follow") is rendered in web/templates/search_description.hbs via the is_operator block.

3.7 Real-Time Event Flow

When a user follows or unfollows someone:

- The action layer calls send_event with type: "followed_users" and the updated followed_users list.
- Tornado pushes the event to all active clients for that user.
- The client-side event handler calls followed_users_data.set_followed_users(event.followed_users), keeping the in-memory set in sync.

When a message is sent:

- get_followers_who_can_see_stream_message(realm_id, sender_id, stream_id, topic_name) computes the set of users who (a) follow the sender, (b) have enable_followed_user_push_notifications=True, and (c) can see the message.
- These user IDs are passed into the push notification queue as followed_user_push_user_ids.

3.8 Frontend Module: followed_users_data.ts

```
// web/src/followed_users_data.ts
const followed_users = new Set<number>();
```

Public API:

Function	Description
initialize(data)	Called on page load with the server-provided followed_users array
set_followed_users(list)	Replaces the entire set — used on real-time events
add_followed_user(id)	Adds a single user ID
remove_followed_user(id)	Removes a single user ID (idempotent)
is_user_followed(id)	Returns true if the ID is in the set

3.9 Developer Workflow

Setup

```
git clone https://github.com/zulip/zulip
cd zulip && ./tools/provision
vagrant up && vagrant ssh
```

```
./tools/run-dev    # dev server at http://localhost:9991
```

Running Tests

```
./tools/test-backend zerver/tests/test_followed_users.py
./tools/test-js-with-node web/tests/followed_users_data.test.cjs
./tools/test-js-with-node web/tests/filter.test.cjs
./tools/lint zerver/tests/test_followed_users.py
```

Key Branches

- main — upstream Zulip
- follow-user — primary feature implementation
- demo-bugfix — three targeted fixes (search description, tooltip template, notification print)

4. Test Plan and Results

4.1 Backend Tests (zerver/tests/test_followed_users.py)

The backend test class FollowedUsersTests(ZulipTestCase) covers the full API surface. Hamlet is the follower; Cordelia is the followed user — mirroring the convention used in test_muted_users.py.

Test Method	What It Verifies	Status
test_get_user_follows	GET endpoint returns empty list, then correct entry after follow	Drafted
test_follow_self	Following yourself returns 400 with appropriate error	Drafted
test_follow_bot	Bots can be followed and unfollowed	Drafted
test_follow_already_followed	Duplicate follow returns 400	Drafted
test_follow_valid_data	Follow succeeds, DB row created, timestamp correct	Drafted
test_follow_deactivated_user	Deactivated users can still be followed	Drafted
test_unfollow_before_following	Unfollow without follow returns 400	Drafted
test_unfollow_valid_data	Unfollow removes DB row, get_user_follows updates	Drafted
test_unfollow_deactivated_user	Deactivated users can be unfollowed	Drafted
test_get_following_users_cache	Cache populated then invalidated on follow/unfollow	Drafted
test_follow_sends_event	Follow action emits followed_users event to correct users	Drafted
test_unfollow_sends_event	Unfollow action emits event with empty followed_users list	Drafted
test_unfollow_is_idempotent_at_action_layer	do_unfollow_user does not raise on non-existent follow	Drafted

4.2 Frontend Tests

web/tests/followed_users_data.test.cjs

Test	What It Verifies	Status
edge_cases	is_user_followed(undefined) and (0) return false	Drafted
add_and_remove_follows	add/remove are idempotent; is_user_followed reflects state	Drafted
set_followed_users	Bulk set replaces state; empty set clears all	Drafted

initialize	initialize() populates correct IDs from server payload	Drafted

web/tests/filter.test.cjs additions

- Predicate test: is:followed-user returns true/false based on followed_users_data.is_user_followed mock.
- Metadata test: correct title, icon, description, redirect URL, and help link for is:followed-user.
- can_newly_match_moved_messages: returns true for is:followed-user and negated variant.

4.3 Known Limitations and Future Work

Limitation	Recommended Next Step
No UI toggle for enable_followed_user_push_notifications	Add a checkbox to Settings > Notifications
No ability to view who you are following from profile page	Add "Following" tab to user settings
Tests drafted but not fully passing (need validation run)	Run ./tools/test-backend and fix remaining assertion errors
Help center article is a stub	Write full MDX article in starlight_help/src/content/docs/
API changelog entry uses placeholder feature level	Run tools/create-api-changelog to finalize before merge
demo-bugfix PR not yet merged to main	Merge after gh auth login and PR approval

5. License Agreement

All code contributed as part of this project is submitted to the Zulip open-source project under the Apache License, Version 2.0. This is consistent with Zulip's existing license and with Cornell University's open-source contribution policy for CS 5150 course projects.

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